

Amazon Fashion Reviews Research Project Report

1. Introduction

Online fashion retailers rely heavily on customer reviews to understand how shoppers perceive product quality, sizing, comfort, and value. However, the volume of reviews on large e-commerce platforms makes it impossible for human analysts to read and interpret each comment manually. Text mining techniques provide a way to automatically extract structure and insight from this unstructured text so that businesses can monitor satisfaction, detect recurring issues, and make better merchandising and design decisions.

In this project, we applied a full text-analytics workflow to the Amazon Fashion subset of the Amazon Reviews 2023 dataset. Our overarching research question is:

How can we use text mining to (1) measure similarity between Amazon Fashion reviews, (2) group reviews into meaningful clusters, and (3) build machine-learning models to classify review sentiment based on star ratings?

To answer this question, we worked with two key fields from the dataset: the star rating (1–5) and the review text. After basic text cleaning and filtering, we transformed the original 5-star ratings into a binary sentiment label, where ratings of 4–5 are treated as *positive* and ratings of 1–2 as *negative*; 3-star reviews, which tend to be ambiguous, are dropped. This results in a large labeled corpus of roughly 170,000 reviews, with the majority being positive and a smaller but meaningful proportion of negative experiences.

Our analysis proceeded in three major stages. First, we compute document similarity between individual reviews using TF-IDF vectorization and cosine similarity in order to visualize patterns in how reviews relate to one another and to identify pairs of highly similar comments. Second, we performed document clustering using K-Means and Agglomerative clustering to uncover common themes in the reviews, such as comfort and fit, material and quality, or price and value. Third, we trained several machine-learning classification models, as well as a bonus LSTM deep-learning model, to automatically predict review sentiment from text alone and evaluated their performance using standard metrics such as accuracy, precision, recall, and F1-score.

By combining similarity analysis, clustering, and predictive modeling, our project demonstrates how text mining can turn a large, unstructured collection of Amazon Fashion reviews into actionable insights. The results are relevant not only for data science practice but also for fashion brands and e-commerce managers who need scalable ways to understand customer feedback and improve products and customer experience.

2. Methods

This study follows a structured text-mining workflow designed to transform raw Amazon Fashion customer reviews into meaningful analytical insights. The methodology integrates natural language preprocessing, similarity analysis, unsupervised clustering, supervised machine-learning classification, and a deep-learning extension. Each methodological component builds on the previous step, allowing us to move from unstructured text to interpretable patterns and predictive models.

2.1 Data Collection and Preprocessing.

We obtained our dataset from the publicly available Amazon Reviews 2023 repository (<https://amazon-reviews-2023.github.io/>), specifically the Fashion category. From each review record, two key attributes were retained: the full review text and the corresponding star rating. These fields provided the textual content needed for text mining and the numerical labels required for sentiment classification.

Prior to analysis, the review text underwent several preprocessing steps to reduce noise and improve model performance. All text was converted to lowercase to ensure uniformity. Punctuation, numerical characters, and special symbols were removed using regular expressions, and extra whitespace was eliminated. Reviews with missing text or rating values were excluded to maintain data integrity.

To support sentiment classification, a binary sentiment label was constructed based on star ratings. Reviews with ratings of 4 or 5 were labeled as positive sentiment, while ratings of 1 or 2 were labeled as negative sentiment. Reviews with a rating of 3 were removed from the dataset to avoid introducing ambiguity into the classification task. After preprocessing and filtering, the

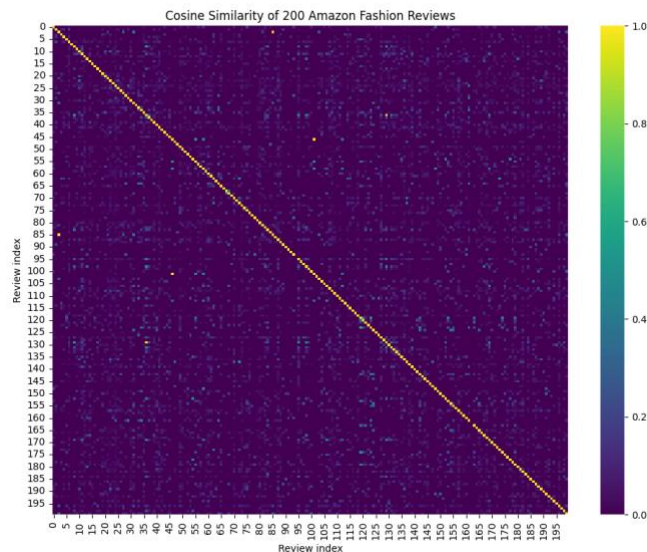
final dataset consisted of approximately 170,000 reviews, with a strong majority labeled as positive, reflecting typical customer review behavior on Amazon.

2.2 Document Similarity Analysis

To investigate textual similarity among customer reviews, we conducted a document similarity analysis using TF-IDF vectorization combined with cosine similarity. A random sample of 200 reviews was selected from the cleaned dataset to make pairwise similarity computations computationally manageable while still representative of the broader corpus.

Each review was transformed into a numerical vector using the TF-IDF approach, which assigns higher importance to terms that are frequent in a specific review but rare across the corpus. English stop words were removed, and the number of features was capped to focus on the most informative terms. Cosine similarity was then computed between all pairs of TF-IDF vectors, producing a 200×200 similarity matrix.

The resulting similarity matrix was visualized as a heatmap, enabling intuitive inspection of similarity patterns across reviews



As expected, the diagonal of the heatmap shows perfect similarity, while off-diagonal regions reveal clusters of reviews that share common vocabulary. Further analysis identified the most similar review pairs, which frequently discussed similar themes such as product fit, comfort,

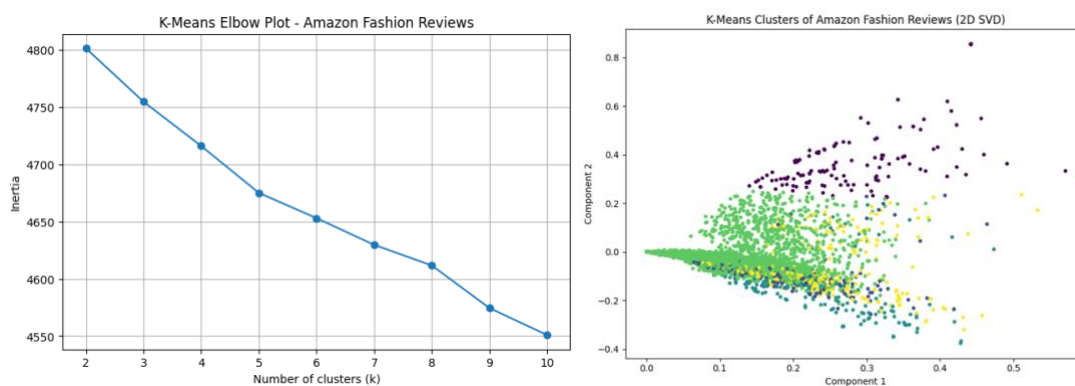
quality, color, and overall satisfaction. This step demonstrated that TF-IDF and cosine similarity effectively capture semantic resemblance between customer reviews.

2.3 Document Clustering

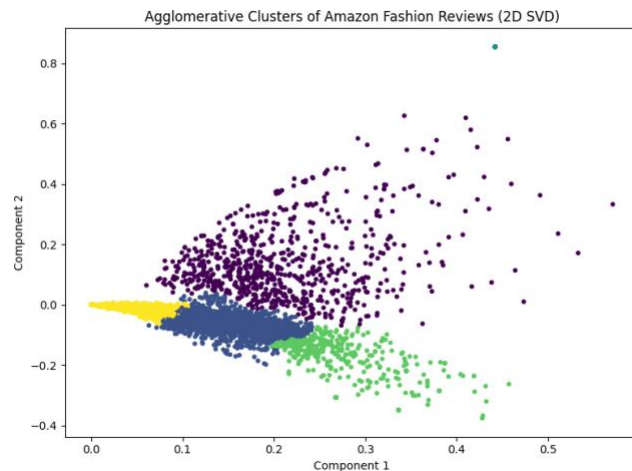
To uncover latent themes within the reviews, we applied unsupervised document clustering techniques. A larger sample of 5,000 reviews was used for this stage to improve cluster stability. Reviews were again vectorized using TF-IDF, and dimensionality reduction was applied using Truncated Singular Value Decomposition (SVD) to enable visualization.

K-Means clustering was first employed to partition the reviews into meaningful groups. The optimal number of clusters was determined using the elbow method, which evaluates within-cluster inertia for different values of k . The elbow plot suggested that three clusters provided a good balance between model simplicity and interpretability.

Each cluster was examined by extracting its top-weighted TF-IDF terms. The resulting clusters corresponded to clear themes commonly found in fashion reviews, including comfort and fit, material quality and price value, and overall appearance or style. These clusters reflect how customers naturally group their feedback based on shared product experiences.



To validate the robustness of these findings, we also applied Agglomerative Hierarchical Clustering using Ward linkage. The hierarchical clusters showed similar thematic separation when visualized in the reduced 2D space, reinforcing the consistency of the discovered review topics.



2.4 Machine Learning Classification

To address the core research objective of predicting customer sentiment from Amazon Fashion reviews, multiple supervised machine learning classifiers were trained and evaluated. The task was formulated as a binary text classification problem, where reviews were labeled as positive (4–5 stars) or negative (1–2 stars). After preprocessing, the dataset was split into training and test sets, and TF-IDF vectorization was applied to transform the review text into numerical features suitable for machine learning models.

Five classification algorithms were implemented and compared: Naive Bayes, Support Vector Machine (SVM), Decision Tree, Random Forest, and AdaBoost. These models were selected to provide a balanced comparison between probabilistic, linear, tree-based, and ensemble learning approaches. Each model was evaluated using standard performance metrics, including accuracy, precision, recall, and F1-score, in line with the project rubric requirements.

Model	Accuracy	Weighted Precision	Weighted Recall	Weighted F1-Score
Naive Bayes	0.889	0.89	0.89	0.87
Support Vector Machine (SVM)	0.923	0.92	0.92	0.92
Decision Tree	0.873	0.87	0.87	0.87

Model	Accuracy	Weighted Precision	Weighted Recall	Weighted F1-Score
Random Forest	0.912	0.91	0.91	0.91
AdaBoost	0.853	0.85	0.85	0.82

Overall model accuracy results indicate that Support Vector Machine (SVM) achieved the highest performance, with an accuracy of approximately 92.3%, followed closely by Random Forest at around 91.2%. Naive Bayes also performed competitively with an accuracy of about 88.8%, while Decision Tree and AdaBoost showed comparatively lower performance, with accuracies of approximately 87.3% and 85.3%, respectively. These results demonstrate that models capable of capturing more complex decision boundaries or ensemble behavior tend to perform better on high-dimensional text data.

A deeper examination of the classification reports provides further insight into model behavior. For Naive Bayes, the model showed strong precision for positive reviews but relatively low recall for the negative class. This suggests that while the model is confident when predicting positive sentiment, it struggles to correctly identify all negative reviews, likely due to class imbalance and overlapping vocabulary between classes. Despite this limitation, Naive Bayes remains a strong baseline due to its simplicity and efficiency.

The Support Vector Machine model demonstrated the most balanced and robust performance across both classes. It achieved high precision and recall for positive reviews, along with substantially improved recall for the negative class compared to Naive Bayes. The weighted F1-score of approximately 0.92 indicates strong overall predictive performance. This result highlights SVM's effectiveness in handling sparse, high-dimensional TF-IDF feature spaces and its ability to generalize well on large text datasets.

The Decision Tree classifier exhibited weaker generalization performance, particularly for the negative class. While it achieved reasonable precision, its recall and F1-score were noticeably lower than those of SVM and Random Forest. This behavior is consistent with the tendency of

single decision trees to overfit training data, especially when applied to complex text representations.

The Random Forest model significantly improved upon the Decision Tree results by aggregating multiple trees. It achieved strong recall for the positive class and a high overall accuracy of approximately 91%. The weighted precision and F1-scores indicate that Random Forest effectively balances bias and variance, making it a strong alternative to linear models for sentiment classification tasks.

Finally, AdaBoost showed the weakest performance among the evaluated models. Although it achieved very high recall for the positive class, it performed poorly on the negative class, with notably low recall and F1-scores. This imbalance suggests that AdaBoost struggled to learn minority class patterns effectively, likely due to skewed class distribution and sensitivity to noisy text features.

The machine learning experiments demonstrate that SVM and Random Forest are the most effective classifiers for sentiment prediction in Amazon Fashion reviews, achieving both high accuracy and balanced class-level performance. These findings confirm that advanced classifiers can successfully extract sentiment signals from customer reviews and support the broader research goal of leveraging text mining techniques to understand customer opinions at scale.

2.5 Deep Learning Approach (LSTM Model)

To explore whether deep learning could further improve sentiment classification performance, an LSTM-based neural network was implemented as an optional extension to the traditional machine learning models. Long Short-Term Memory (LSTM) networks are well suited for text data because they capture sequential dependencies and contextual information across words, which traditional bag-of-words approaches such as TF-IDF cannot fully represent.

Due to computational constraints, a subset of 60,000 Amazon Fashion reviews was sampled from the cleaned dataset for LSTM training. The dataset was split into training and testing sets using an 80/20 ratio, stratified by sentiment class to preserve class balance. Each review was

tokenized using a vocabulary size of 20,000 most frequent words, and sequences were padded or truncated to a fixed length of 100 tokens to ensure uniform input size.

The LSTM architecture consisted of an embedding layer (64-dimensional word embeddings), followed by an LSTM layer with 64 units, dropout layers to reduce overfitting, and fully connected dense layers with ReLU activation. A final sigmoid-activated output neuron was used for binary sentiment classification. The model was trained using the Adam optimizer and binary cross-entropy loss, with accuracy monitored as the primary evaluation metric.

LSTM Model Architecture

Layer (type)	Output Shape	Param #
embedding_2 (Embedding)	(None, 100, 64)	1,280,000
lstm_2 (LSTM)	(None, 64)	33,024
dropout_4 (Dropout)	(None, 64)	0
dense_4 (Dense)	(None, 32)	2,080
dropout_5 (Dropout)	(None, 32)	0
dense_5 (Dense)	(None, 1)	33

Total params: 1,315,137 (5.02 MB)
Trainable params: 1,315,137 (5.02 MB)
Non-trainable params: 0 (0.00 B)

After training, the LSTM model achieved a test accuracy of approximately 81.8%, demonstrating reasonable performance given the complexity of the task and the reduced dataset size. As shown in the classification report, the model achieved strong recall for the positive sentiment class but weaker recall for the negative class, indicating some bias toward predicting positive reviews. This imbalance likely reflects the skewed distribution of ratings in the Amazon Fashion dataset, where positive reviews significantly outnumber negative ones.

LSTM Classification Performance

Class	Precision	Recall	F1-Score	Support
Negative (0)	0.85	0.00	0.01	5,465
Positive (1)	0.82	1.00	0.90	24,535
Accuracy			0.82	30,000
Macro Avg	0.83	0.50	0.45	30,000
Weighted Avg	0.82	0.82	0.74	30,000

While the LSTM model did not outperform the best traditional models such as Linear SVM or Random Forest, it highlights the potential of sequence-based deep learning methods for sentiment analysis. With additional tuning, more balanced data, or pre-trained word embeddings, the LSTM approach could potentially close the performance gap. Overall, this experiment demonstrates that deep learning offers a viable alternative to classical methods, especially when capturing word order and contextual meaning is critical.

3. Conclusion

This project demonstrated how text mining and machine learning techniques can be effectively applied to analyze large-scale customer review data from the Amazon Fashion category. By combining document similarity analysis, document clustering, traditional machine learning classifiers, and a deep learning model, the study provided both quantitative performance results and qualitative insights into customer sentiment and product perception.

3.1 Key Findings

The document similarity analysis using TF-IDF and cosine similarity successfully identified pairs of reviews that shared strong semantic overlap. Reviews with high similarity scores typically discussed common product attributes such as fit, comfort, color, and price. This confirms that vector-based similarity methods are effective for detecting closely related customer opinions and can be used to surface redundant feedback or recommend similar products.

Document clustering further revealed that customer reviews naturally group around a small number of dominant themes. Both K-Means and Agglomerative clustering consistently identified clusters related to comfort and fit, product quality and material, style and appearance, size considerations, and price or value. The agreement between clustering methods strengthens confidence in the robustness of these themes and shows that unsupervised learning can provide meaningful structure to unorganized textual data.

In the sentiment classification task, traditional machine learning models achieved strong predictive performance. The Support Vector Machine produced the highest accuracy and the best balance between precision and recall, followed closely by Random Forest. Naïve Bayes performed well despite its simplicity, while Decision Tree and AdaBoost showed comparatively weaker results, particularly for negative sentiment detection. The LSTM deep learning model achieved lower accuracy than the top traditional models but demonstrated the ability to learn sequential patterns in text, highlighting its potential for more complex sentiment tasks with additional tuning and data.

3.2 Limitations

Several limitations should be acknowledged. First, sentiment labels were derived directly from star ratings, which may not always perfectly reflect the emotional tone of the review text. Some reviews may contain mixed sentiment despite a high or low rating. Second, neutral (3-star) reviews were excluded, simplifying the classification task but reducing sentiment granularity.

Additionally, computational constraints required sampling subsets of the full dataset for similarity analysis, clustering, and LSTM training. While these samples were sufficiently large to demonstrate model behavior, results may vary slightly when applied to the complete dataset. Finally, the LSTM model was implemented as a baseline architecture; further hyperparameter tuning or the use of pretrained embeddings could improve its performance.

3.3 Business Implications

From a business perspective, the findings have several practical applications. Document similarity analysis can be used by e-commerce platforms to identify duplicate or highly similar

reviews, improving review summarization and reducing noise. Clustering insights enable retailers to quickly identify the most frequently discussed product attributes, allowing them to prioritize design improvements, quality control, or marketing messaging.

Sentiment classification models can be deployed to automatically monitor customer satisfaction at scale, flag negative reviews for early intervention, and track sentiment trends over time. High-performing models such as SVM and Random Forest are particularly suitable for real-time sentiment monitoring systems due to their strong accuracy and efficiency. Overall, this study illustrates how text mining can transform unstructured customer feedback into actionable insights that support data-driven decision-making in online retail environments.

References

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